

Applied Chemistry

Unit-1

Fuels

BIT DURG



Department of Applied Chemistry

Bhilai Institute of Technology, Durg (C.G.)

B.Tech. First / Second Semester, BIT Durg, Applied Chemistry; Unit-1

SYLLABUS

UNIT – I : Fuels

CO1

Solid Fuel: Classification, Characteristics, Calorific value, Types, Dulong's Formula, Bomb calorimeter (numericals), Analysis of Coal: Proximate and Ultimate analysis, Flue gas analysis, Significance, Carbonization of Coal, Manufacture of metallurgical coke by Otto Hoffman's byproduct oven, Numerical related to Dulong's Formula, Bomb calorimeter & Combustion (by weight & by volume). Liquid fuel: Petrol knocking, Antiknocking, Octane number, Diesel knocking, Cetane number. **[8Hrs]**

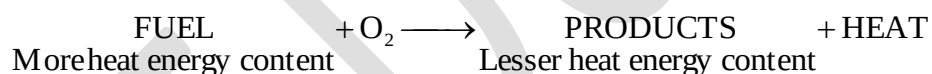
Course Outcome:

CO1: Explain the quality, composition of fuel and solve the related problems.

UNIT-1

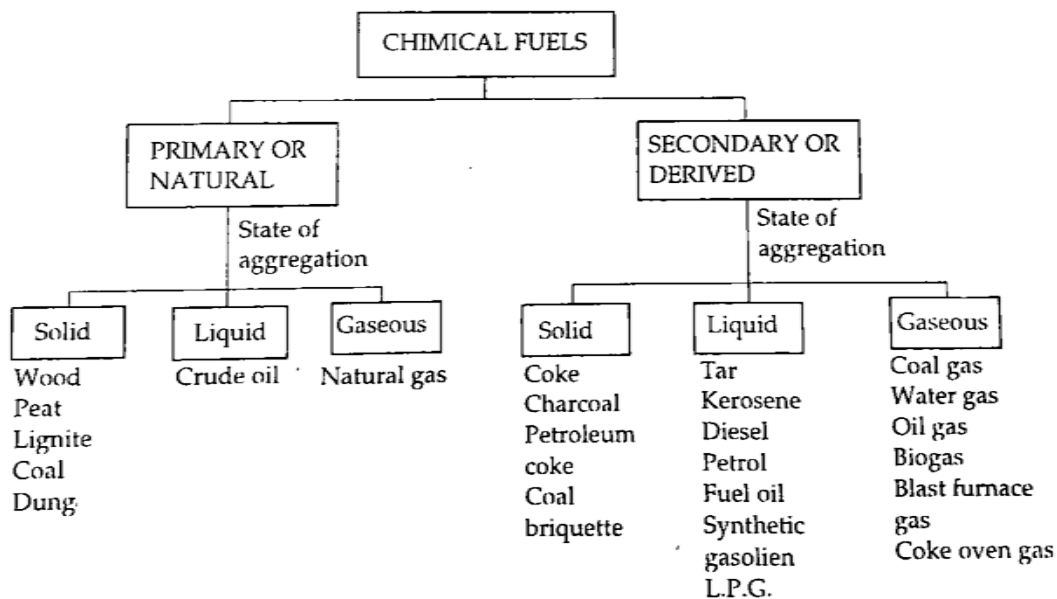
FUELS

Solid Fuel: Fuel is a combustible substance, containing carbon as main constituent, which on proper burning gives large amount of heat, which can be used economically for domestic and industrial purposes. Wood, charcoal, coal, kerosene, petrol, diesel, producer gas, oil gas, etc. are some of the fuels.



The primacy or main source of fuels are coals and petroleum oils. These are stored fuels available in earth's crust and are, generally, called 'fossil fuels'.

Classification of Fuels: The fossil fuels have been classified according to their: (1) occurrence (and preparation), and (2) the state of aggregation. According to the first classification, we have: (a) natural or primary fuels, which are found in nature as such, e.g., wood, peat, coal, petroleum, natural gas, etc.; (b) artificial or secondary fuels are those which are prepared from the primary fuels. For example, charcoal, coke, kerosene oil, diesel oil, petrol, coal gas, oil gas, producer gas, blast furnace gas, etc.



The second classification is based upon their state of aggregation like: (a) solid fuels; (b) liquid fuels, and (c) gaseous fuels.

Calorific Value: Calorific value of a fuel is “the total quantity of heat liberated, when a unit mass (or volume) of the fuel is burnt completely.”

Higher or gross calorific value: Higher calorific value (HCV) is “the total amount of heat produced, when unit mass / volume of the fuel have been burnt completely and the products of combustion have been cooled to room temperature” (i.e., 15°C or 60°F).

Lower or net calorific value: Lower calorific value (LCV) is “the net heat produced, when unit mass / volume of the fuel is burnt completely and the products are permitted to escape”. Alternatively, net or lower calorific value (LCV)

$$= \text{HCV} - \text{Latent heat of water vapour formed}$$

$$= \text{HCV} - \text{Mass of hydrogen} \times 9 \times \text{Latent heat of steam}$$

Because 1 part by mass of hydrogen produces 9 parts by mass of water. The latent heat of steam is 587 kcal / kg or 1,060 Th.U. / lb of water vapour formed at room temperature (i.e., 15°C).

Units of calorific value: The calorific value is, generally, expressed in calorie / gram (cal / g) or kilocalorie / kg (kcal / kg) or British thermal unit / lb (B.Th.U / lb) in case of solid or liquid fuel.

Characteristics of a Good Fuel:

1. High calorific value: A fuel should possess high calorific value, since the amount of heat liberated and temperature attained thereby depends upon the calorific value of fuel.

2. Moderate ignition temperature: Ignition temperature is the lowest temperature to which the fuel must be pre-heated so that it starts burning smoothly. Low ignition temperature is dangerous for storage and transport of fuel, since it can cause fire hazards. High ignition temperature causes difficulty in kindling for igniting) the fuel, but the fuel is safe during storage, handling and transport. Hence, an ideal fuel should have “moderate” ignition temperature.

3. Low moisture content: The moisture content of the fuel reduces the heating value and involves in a loss of money, because It is paid for at the same rate as the fuel. Hence, fuel should have low moisture content. 5 -10 % of moisture is desirable for proper fuel bed preparation and lesser fly ash generation.

4. Low non-combustible matter content: After combustion, the non-combustible matter remains, generally, in the form of ash or clinker. The non-combustible matter also reduces the heating value, besides additional cost of storage, handling and disposal of the waste products produced. Each per cent of non-combustible matter in fuel means a heat loss of about 1.5%. Hence, a fuel should have low content of non-combustible matter.

5. Moderate velocity of combustion: If the rate of combustion is low, then the required high temperature may not be possible, because a part of the heat liberated may get radiated, instead of raising the temperature. On the other hand, too high combustion rates are also not required.

6. Products of combustion should not be harmful: Fuel, on burning, should not give out objectionable and harmful gases. In other words, the gaseous products of combustion should not pollute the atmosphere. CO, SO₂, H₂S, PH₃, etc., are some of the harmful gases.

7. Low cost: A good fuel should be readily available in bulk at a cheap rate.

8. Easy to transport: Fuel must be easy to handle, store and transport at a low cost. Solid and liquid fuels can easily be transported from one place to another. On the other hand, transportation of gaseous fuels is costly and can even cause fire hazards.

9. Combustion should be easily controllable, i.e., combustion of the fuel should be easy to start or stop, when required.

Dulong’s Formula:

Theoretical Calculation of Calorific Value of a Fuel: The calorific value of fuel can be approximately computed by noting the amounts of the constituents of the fuel. The higher calorific values of some of the chief combustible constituents of fuel are tabulated below:

Table 1. Calorific values of fuel constituents

Constituent	Hydrogen	Carbon	Sulphur
HCV (kcal / kg)	34,500	8,080	2,240

The oxygen, if present in the fuel, is assumed to be present in combined form with hydrogen, i.e., in the form of fixed hydrogen [H₂O]. So, the amount of hydrogen available for combustion

$$= \text{Total mass of hydrogen in fuel} - \text{Fixed hydrogen}$$

$$= \text{Total mass of hydrogen in fuel} - (1/8) \text{ Mass of oxygen in the fuel}$$

($\therefore$ 8 parts of oxygen combine with one part of hydrogen to form H₂O)

Dulong's formula for calorific value from the chemical composition of fuel is:

$$\text{HCV} = \frac{1}{100} \left[8,080C + 34,500 \left(H - \frac{O}{8} \right) + 2,240S \right] \text{ kcal / kg}$$

Where, C, H, O, and S are the percentages of carbon, hydrogen, oxygen and sulphur in the fuel respectively. In this formula, oxygen is assumed to be present in combination with hydrogen as water, and

$$\text{LCV} = \left[\text{HCV} - \frac{9}{100} H \times 587 \right] \text{ kcal / kg} = [\text{HCV} - 0.09H \times 587] \text{ kcal / kg}$$

This is based on the fact that 1 part of H by mass gives 9 parts of H₂O, and latent heat of steam is 587 kcal / kg.

Analysis of Coal: In order to assess the quality of coal, the following two type of analysis are made:

(A) Proximate analysis involves in the following determinations:

1. Moisture: About 1 g of finely powered air-dried coal sample is weighed in a crucible. The crucible is placed inside an electric hot air-oven, maintained at 105°-110°C. The crucible is allowed to remain in oven for 1 hour and then taken out (with the help of a pair of tongs), cooled in a desiccators and weighed. Loss in weight is reported as moisture (on percentage-basis).

2. Volatile matter: The dried sample of coal left in the crucible in (1) is then covered with a lid and placed in an electric furnace (muffle furnace), maintained at 925° ± 20°C. The crucible is taken out of the oven after 7 minutes of heating. The crucible is cooled first in air, then inside a desiccator and weighed again. Loss in weight is reported as volatile matter on percentage-basis.

$$\text{Percentage of volatile matter} = \frac{\text{Loss in weight due to removal of volatile matter}}{\text{Weight of coal sample taken}} \times 100$$

3. Ash: The residual coal in the crucible in (2) is then heated without lid in a muffle furnace at 700 ± 50°C for 1/2 hour. The crucible is then taken out, cooled first in air, then in desiccators and weighed. Heating, cooling and weighing is repeated, till a constant weight is obtained. The residue is reported as ash on percentage-basis. Thus, percentage of ash

$$= \frac{\text{Weight of ash left}}{\text{Weight of coal taken}} \times 100$$

4. Fixed carbon: Percentage of fixed carbon = 100 – % of (moisture + volatile matter + ash)

Importance of proximate analysis: Proximate analysis provides following valuable in formations in assessing the quality of coal:

1. **Moisture** in coal evaporates during the burning of coal and it takes some of the liberated heat, in the form of latent heat of evaporation. Therefore, moisture lowers the effective calorific value of coal. Moreover, it quenches the fire in the furnace. Hence, lesser the moisture content, better the quality of coal as a fuel. However, presence of moisture, up to 10%, produces a more uniform fuel-bed and less of “fly-ash”.

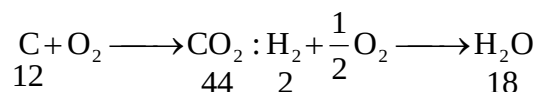
2. **Volatile matter:** A high volatile matter content means that a high proportion of fuel will distill over as gas or vapour, a large proportion of which escapes unburnt, So, higher volatile content in coal is undesirable. A high volatile matter containing coal burns with a long flame, high smoke and has low calorific value. Hence, lesser the volatile matter, better the rank of the coal.

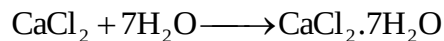
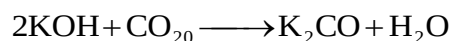
3. **Ash** is a useless, non-combustible matter, which reduces the calorific value of coal. Moreover, ash causes the hindrance to the flow of air and heat, thereby lowering the temperature. Also, it often causes trouble during firing by forming clinkers (i.e., fused ash lumps), which block the interspaces of the grate, on which coal is being lower the ash content, better the quality of coal. The presence of ash (similar to moisture) also increases transporting, handling and storage costs. It also involves additional cost in ash disposal.

4. **Fixed carbon:** Higher the percentage of fixed carbon, greater is it's calorific and betters the quality coal. Greater the percentage of fixed carbon, smaller is the percentage of volatile matter. This also represents the quantity of carbon (in coal) that can be burnt by a primary current of air drawn through the hot bed of a fuel. Hence, high percentage of fixed carbon is desirable. The percentage of fixed carbon helps in designing the furnace and the shape of the fire-box, because it is the fixed carbon that burns in the solid state.

(B) Ultimate analysis involves in the following determinations:

1. **Carbon and hydrogen:** About 1-2 g of accurately weighed coal sample is burnt in a current of oxygen in a combustion apparatus. C and H of the coal are converted into CO₂ and H₂O respectively. The gaseous products of combustion are absorbed respectively in KOH and CaCl₂ tubes of known weights. The increase in weights of these are then determined.





$$\therefore \text{Percentage of C} = \frac{\text{Increase in weight of KOH tube} \times 12 \times 100}{\text{Weight of coal sample taken} \times 44}$$

$$\text{and percentage of H} = \frac{\text{Increase in weight of CaCl}_2 \text{ tube} \times 2 \times 100}{\text{Weight of coal sample taken} \times 18}$$

2. Nitrogen: About 1 g of accurately weighed powdered coal is heated with concentrated H_2SO_4 along-with K_2SO_4 (catalyst) in a long-necked flask (called Kjeldahl's flask). After the solution becomes clear, it is treated with excess of KOH and the liberated ammonia is distilled over and absorbed in a known volume of standard acid solution. The unused acid is then determined by back titration with standard NaOH solution. From the volume of acid used by ammonia liberated, the percentage of N in coal is calculated as follows:

$$\text{Percentage of N} = \frac{\text{Volume of acid used} \times \text{Normality} \times 1.4}{\text{Weight of coal taken}}$$

3. Sulphur: is determined from the washings obtained from the known mass of coal, used in a bomb calorimeter for determination of a calorific value. During this determination, S is converted into sulphate. The washings are treated with barium chloride solution, when barium sulphate is precipitated. This precipitate is filtered, washed and heated to constant weight.

$$\text{Percentage of S} = \frac{\text{Weight of BaSO}_4 \text{ obtained} \times 32 \times 100}{\text{Weight of coal sample taken in bomb} \times 233}$$

4. Ash determination is carried out as in proximate analysis.

5. Oxygen: It is obtained by difference.

$$\text{Percentage of O} = 100 - \text{Percentage of (C + H + S + N + ash)}$$

Importance of ultimate analysis:

1. Carbon and hydrogen: Greater the percentage of carbon and hydrogen better is the coal in quality and calorific value. However, hydrogen is mostly associated with the volatile matter and hence, it affects the use to which the coal is put.

2. Nitrogen has no calorific value and hence, its presence in coal is undesirable. Thus, a good quality coal should have very little nitrogen content.

3. Sulphur, although contributes to the heating value of coal, yet on combustion produces acids (SO_2 and SO_3), which have harmful effects of corroding the equipments and also cause atmospheric pollution. Sulphur is, usually, present to the extent of 0.5 to 3.0% and derived from ores like iron pyrites, gypsum, etc., mines along with the coal.

4. Oxygen content decreases the calorific value of coal. High oxygen-content coals are characterized by high indexed moisture, low calorific value, and low coking power. Moreover, oxygen is in combined form with hydrogen in coal and thus, hydrogen available for combustion is lesser than actual one. An increase in 1% oxygen content decreases the calorific value by about 1.7% and hence, oxygen is undesirable. Thus, a good quality coal should have low percentage of oxygen.

Carbonization of coal: These are two types of carbonization of coal:

1. Low-temperature carbonization: In this process, the heating of coal is carried out at $500 - 700^\circ\text{C}$. The yield of coke is about 75 – 80% and it contains about 5 – 15% volatile matter. It is not mechanically strong, so it cannot be used as a metallurgical coke. However, it burns easily giving practically a smokeless, hot and radiant fire. Hence, it is suitable for domestic purposes. The by-product gas produced (about $130 - 150 \text{ m}^3 / \text{tonne}$) by this process is richer in heating value (about $6,500 - 9,500 \text{ kcal} / \text{m}^3$) and is, therefore, a more valuable gaseous fuel.

2. High-temperature carbonization is carried out at $900 - 1,200^\circ\text{C}$ with the object of producing coke of the right porosity, hardness, purity strength, etc., so that it can be used in metallurgy. Nearly all the volatile matter of coal is driven off and the yield of coke is about 65 - 75%, containing only 1 - 3% volatile matter. The by-product gas produced is high in volume (about $300 - 390 \text{ m}^3 / \text{tonne}$), but its calorific value is low (about $5,400 - 6,000 \text{ kcal} / \text{m}^3$).

Table 2. Low and high-temperature carbonizations

S. No.	Low-temperature carbonization	High-temperature carbonization
1.	Heating temperature: About $500 - 700^\circ\text{C}$.	$900 - 1,200^\circ\text{C}$
2.	Yield of coke: 75 - 80%.	65 – 75%
3.	Percentage of volatile matter in coke produced: It contains about 5 - 15% volatile matter.	It contains about 1 – 3% volatile matter.
4.	Mechanical strength of coke: It is not mechanically strong.	Coke produced is hard and of good mechanical strength.
5.	Use of coke: Good for domestic purposes.	Suitable for metallurgical purposes.
6.	Quantity of by-product gas produced: About $130 - 150 \text{ m}^3 / \text{tonne}$.	About $300 - 390 \text{ m}^3 / \text{tonne}$.
7.	Calorific value of gas: About $6,500 - 9,500 \text{ kcal} / \text{m}^3$.	About $5,400 - 6,000 \text{ kcal} / \text{m}^3$.
8.	Percentage of aromatic hydro carbons in the gas: Lower.	Higher.

S. No.	Low-temperature carbonization	High-temperature carbonization
9.	Percentage of straight-chain hydrocarbons in the gas: Higher.	Lower.
10.	Hardness of coke produced: Soft.	Hard.
11.	Smoke produced on burning coke: Smokeless.	Smoky.
12.	Percentage of aromatic hydrocarbons in tar: Smaller.	Higher.

Manufacture of Metallurgical Coke:

Otto Hoffman's by-product oven: In order to: (i) increase the thermal efficiency of the carbonization process, and (ii) recover valuable by-product (like coal gas, ammonia, benzol oil, tar, etc.), Otto Hoffman developed modern by-product coke oven which, unlike beehive oven, is heated externally by a portion of coal gas produced during the process itself or by producer gas or by blast furnace gas. Moreover, the heating is done on the basis of "regenerative system of heat economy", i.e., utilizing the waste flue gases for heating the checker-work of bricks.

The by-product coke oven consists of number of narrow silica chambers (each about 10 to 12 m long, 3 to 4 m high and 0.40 to 0.45 m wide) erected side-by-side with vertical flues in-between them to form a sort of battery. Each chamber is provided with a charging hole at the top, a gas off-take and a refractory-lined cast iron door at each ends for discharging coke.

A charge consisting of finely crushed coal is introduced through the charging holes at the top of chambers, which are then closed tightly at both ends to prevent any access of air. The coke ovens are heated to 1,200°C by burning gaseous fuel (like producer gas) and usually employing a regenerative principle to achieve as economical heating as possible. The flue gases produced during combustion, before escaping to chimney, pass on their sensible heat to one of the two sets of checker brick-work, until this brick-work has been raised to a temperature of about 1,000°C. The flow of heating gases is then reversed and the inlet gases are passed through the heated checker brick-work, which thus serves to preheat the inlet gases. The flue gases are then allowed to pass through the second set of checker bricks to heat it. This checker-work then serves to preheat the inlet gases, when the flow is again reversed. Thus, this cycle goes on. The heating is actually continued, till the evolution of volatile matter ceases completely. Carbonization of a charge of coal takes about between 11 to 18 hours.

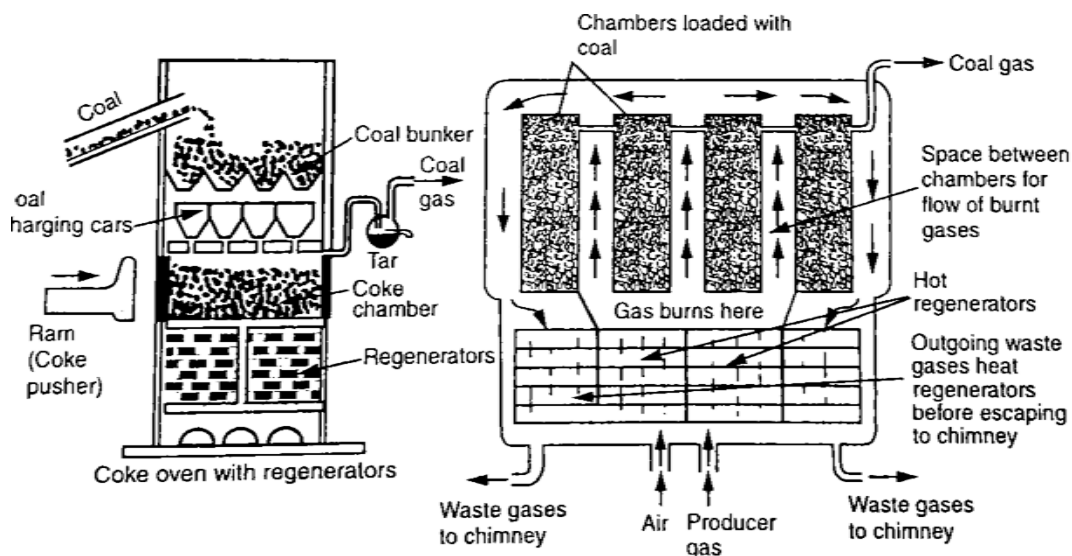


Fig. 1. Otto Hoffman's by-product coke oven with regenerators.

When carbonization is completed, a massive ram pushes the red hot coke into a truck. It is subsequently quenched by a water spray ('wet quenching'). In place of wet quenching, "dry quenching" offers advantages, because the coke produced is more strong, dense, graphitized and non-reactive. In this method, the red hot coke is placed in a chamber and cooled by passing inert gases from boilers (like nitrogen). The heated inert gases are then circulated to boilers, where they generate steam. The coke produced by 'dry quenching' is cheaper, drier and contains lesser dust than 'wet-quenched' coke.

Recovery of by-products: The gas coming out from the oven is known as "coke oven gas" and is mainly composed of ammonia, H_2S , naphthalene, benzene, tar, moisture, etc.

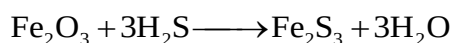
(i) Recovery of tar: The gas is first passed through a tower in which liquor ammonia is sprayed. Here dust and tar get collected in a tank below, which is heated by steam coils to recover back ammonia sprayed. The ammonia is used again.

(ii) Recovery of ammonia: The gases from the chamber are then passed through a tower in which water is sprayed. Here ammonia goes into solution as NH_4OH .

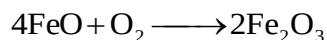
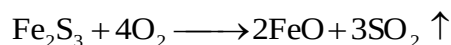
(iii) Recovery of naphthalene: The gases are then passed through another tower in which water at very low temperature is sprayed. Here naphthalene gets condensed.

(iv) Recovery of benzene: The gases are then sprayed with petroleum, when benzene and its homologues are removed.

(v) Recovery of H_2S : The gases are then passed through a purifier, packed with moist Fe_2O_3 . Here H_2S is retained.



After some time, when all Fe_2S_3 is changed into Fe_2O_3 , the purifier is exposed to atmosphere, when Fe_2O_3 is regenerated.



Petroleum:

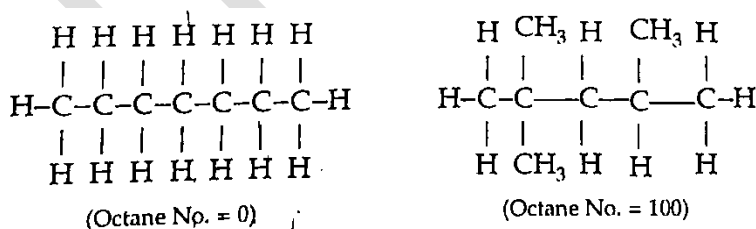
Knocking: In an internal combustion engine, a mixture of gasoline vapour and air is used as a fuel. The ratio of the gaseous volume in the cylinder at the end of the suction-stroke to the volume at the end of compression-stroke of the piston, is known the 'compression ratio'. The efficiency of an internal combustion engine increases with the increase in compression ratio, which is dependent on the nature of the constituents present in the gasoline used. The rate of oxidation becomes so great that the last portion of the fuel-air mixture gets ignited instantaneously, producing an explosive violence, known as 'knocking'. The knocking results in loss of efficiency.

Chemical structure and knocking: The tendency of fuel constituents to knock is in the following order:

Straight-chain paraffins > branched-chain paraffins (i.e., iso-paraffins) > olefins > cycle-paraffins (i.e., naphthalenes) > aromatics.

Thus, olefines of the same carbon-chain length possess better anti-knock properties than the corresponding paraffins and so on.

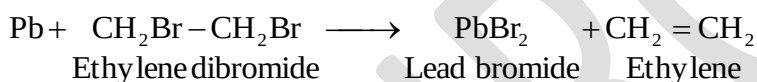
Octane rating (introduced by Edger in 1872): It has been found that n-heptane knocks very badly and hence, its anti:knock value has arbitrarily been given zero. On the other hand, isooctane (2 : 2 : 4-trimethyl pentane),



given very little knocking, so its anti-knock value has been as 100. Thus, octane number (or rating) of a gasoline (or any other internal combustion engine fuel) is the percentage of isooctane in a mixture of isooctane and n-heptane, which matches the fuel under test in knocking characteristics. In this way, an “80-octane” fuel is one which has the same combustion characteristics as a 80 : 20 mixture of isooctane and n-heptane.

Improvement of anti-knock characteristics of a fuel: The octane number of LC. Fuels can be raised by the addition of such extremely poisonous materials as tetra ethyl lead, $(C_2H_5)_4 Pb$ or TEL and diethyl telluride, $(C_2H_5)_2 Te$. In motor spirit (or motor fuel), about 0.5 mL and in aviation fuels, about 1.0 to 1.5 ml of TEL is added per litre of petrol.

According to the most accepted theory, TEL is converted into a cloud of finely divided lead oxide particles in the cylinder and these particles react with any hydrocarbon peroxide molecules formed, thereby slowing down the chain oxidation reaction and thus, decreasing the chances of any early detonation. However, deposit of lead oxide is harmful to the engine life. Consequently, in order to help the simultaneous elimination of lead oxide formed from the engine, a small amount of ethylene dibromide is also added to petrol. Ethylene dibromide removes lead oxide as volatile lead bromide along with the exhaust gases. The presence of sulphur compounds in petrol reduces the effectiveness of the TEL.



Diesel Engine Fuels: In a diesel engine, the fuel is exploded not by a spark, but by the application of heat and pressure. Diesel engine fuels consist of longer chain hydrocarbons than internal combustion engine fuels. The main characteristic of diesel engine fuel is that it should easily ignite below compression temperature, and there should be as short an induction lag as possible. The hydrocarbon molecules in a diesel fuel should be as far as possible the “straight-chain” ones, with a minimum admixture of aromatic and side-chain hydrocarbon molecules.

The suitability of a diesel fuel is determined by its cetane value, which is the percentage of hexadecane in a mixture of hexadecane and 2-methyl naphthalene, which has the same ignition characteristics as the diesel fuel in question. The cetane number of a diesel fuel can be raised by the addition of small quantity of certain “pre-ignition dopes” like ethyl nitrite, isoamyl nitrite, acetone peroxide, etc.

Ignition quality order among hydrocarbon constituents of a diesel fuel is as follows:

n-alkanes > naphthalenes > alkenes > branched alkanes > aromatics.

Thus, hydrocarbons which are poor gasoline fuels are quite good diesel fuels.

Analysis of Flue Gas: In order to have proper control on combustion process, an idea about the complete or incomplete combustion of fuel is made by the analysis of flue gas. Thus, (i) if the flue gases contain considerable amount of carbon monoxide, it indicates that incomplete combustion is occurring (i.e., considerable wastage of fuel is taking place) and it also indicates short supply of oxygen for combustion. On the other hand, (ii) if the flue gases contain a considerable amount of oxygen, it indicates the oxygen supply is in excess, though the combustion may be complete.

The analysis of flue gases is made with the help of Orsat's apparatus. It consists of a water-jacketed measuring burette, connected in series to a set of three absorption bulbs, each through a stop-cock. The other end is provided with a three-way stop-cock, the free end of which is further connected to a U-tube packed with glass wool (for avoiding the incoming of any smoke particles, etc.). The graduated burette is surrounded by a water-jacket to keep the temperature of gas constant during the experiment. The lower end of the burette is connected to a water reservoir by means of a long rubber tubing. The absorption bulbs are usually filled with glass tubes, so that the surface area of contact between the gas and the solution is increased.

The absorption bulbs have solutions for the absorption of CO_2 , O_2 and CO respectively. First bulb has 'potassium hydroxide' solution (250 g KOH in 500 mL of boiled distilled water), and it absorbs only CO_2 . The second bulb has a solution of 'alkaline pyrogalllic acid' (25 g pyrogalllic acid + 200 g KOH in 500 mL of distilled water) and it can absorb CO_2 and O_2 . The third bulb contains 'ammoniacal cuprous chloride' (100 g cuprous chloride + 125 mL liquor ammonia + 375 mL of water) and it can absorb CO_2 , O_2 and CO . Hence, it is necessary that the flue gas is passed first through potassium hydroxide bulb, where CO_2 is absorbed, then through alkaline pyrogalllic acid bulb, when only O_2 will be absorbed (because CO_2 has already been removed) and finally through ammoniacal cuprous chloride bulb, where only CO will be absorbed.

Working: Step-1: To start with, the whole apparatus is thoroughly cleaned, stoppers greased and then tested for air-tightness. The absorption bulbs are filled with their respective solutions to level just below their rubber connections. Their stop-cocks are then closed. The jacket and leveling reservoir are filled with water. The three-way stop-cock is opened to the atmosphere and reservoir is raised, till the burette is completely filled with water and air is excluded from the burette. The three-way stop-cock is now connected to the flue gas supply and the reservoir is lowered to draw apparatus. So the three-way stop-cock is opened to the atmosphere, and the gas expelled out by raising the reservoir. This process of sucking and exhausting of gas is repeated 3-4 times, so as to expel the air from the capillary connecting tubes, etc. Finally, gas is sucked in the burette and the volume of the flue gas is adjusted to 100 mL at atmospheric pressure. For adjusting final volume, the three-way stop-cock is opened to atmosphere and the reservoir is carefully raised, till the level of water in it is the same as in the burette, which stands at 100 mL mark. The three-way stop-cock is then closed.

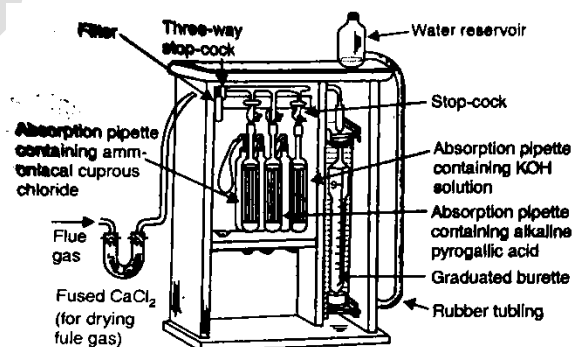


Fig. 2: Orsat's gas analysis apparatus

Step-2: The stopper of the absorption bulb, containing caustic potash solution, is opened and all the gas is forced into this bulb by raising the water reservoir. The gas is again sent to the burette. This process is repeated several times to ensure complete absorption of CO₂ [by KOH solution]. The unabsorbed gas is finally taken back to the burette, till the level of solution in the CO₂ absorption bulb stands at the constant mark and then, its stop-cock is closed. The levels of water in the burette and reservoir are equalised and the volume of residual gas is noted. The decrease in volume gives the volume of CO₂ in 100 mL of the flue gas sample.

Step-3: The volumes of O₂ and CO are similarly determined by passing the remaining gas through alkaline pyrogalllic acid bulb and ammoniacal cuprous chloride bulb respectively. The gas remaining in burette after adsorption of CO₂, O₂, and CO is taken as nitrogen.

Precautions:

1. To start with, the reagents in the absorption bulb 1, 2 and 3 should be brought to the etched mark levels one-by-one by operating the reservoir bottle and the valve of each bulb. Then, their respective valves are closed.
2. All the air from the reservoir bottle is expelled to atmosphere by lifting the reservoir bottle and opening the three-way to atmosphere. The three-way is then connected to the flue gas supply and the reservoir bottle is brought down, until the level in the burette becomes zero (i.e., 100 mL of gas is taken in the burette). The gas in the burette is expelled to the atmosphere to remove any air left in the joints, tubes, etc. This procedure is repeated 2-3 times to ensure a right sample of the gas taken for analysis.
3. It is quite necessary to follow the order of absorbing the gases: CO₂ first, O₂ second, and CO last. This is because the absorbent used for O₂ (i.e., alkaline pyrogalllic acid) can also absorb some CO₂ and the percentage of CO₂ left would be less; while the percentage of O₂ thus-detected would be reported more. The absorbent used for CO₂, however, does not absorb O₂ or CO.
4. The percentage of CO in the flue gas is very small and this should be measured quite carefully.

NUMERICALS

Example-1: Calculate the gross and net calorific value of coal having the following compositions carbon = 85%, hydrogen = 8%, sulphur = 1%, nitrogen = 2%, ash = 4%, latent heat of steam = 587 cal / g.

Solution: Gross calorific value (GCV)

$$= \frac{1}{100} \left[8,080 \times C + 34,500 \left(H - \frac{O}{8} \right) + 2,240 \times S \right] \text{ kcal / kg}$$

$$= \frac{1}{100} \left[8,080 \times 85 + 34,500 \left(8 - \frac{0}{8} \right) + 2,240 \times 1 \right] \text{ kcal / kg}$$

$$= \frac{1}{100} [686,800 + 276,000 + 2,240] \text{ kcal / kg}$$

$$= \frac{1}{100} [965,040] \text{ kcal / kg} = 9,650.4 \text{ kcal / kg}$$

$$\text{Net calorific value (NCV)} = (\text{GCV} - 0.09H \times 587) \text{ kcal / kg}$$

$$= (9,640.4 - 0.09 \times 8 \times 587) \text{ kcal / kg} = 9,227.8 \text{ kcal / kg}$$

Example-2: On burning 0.83 g of a solid fuel in a bomb calorimeter, the temperature of 3,500 g of water increased from 26.5°C to 29.2°C. Water equivalent of calorimeter and latent heat of steam are 385.0 g and 587.0 cal / g respectively. If the fuel contains 0.7% hydrogen, calculate its gross and net calorific value.

Solution: Here weight of fuel (x) = 0.83 g; weight of water (W) = 3,500 g; water equivalent of calorimeter (w) = 385 g; (t₂ - t₁) = 29.2°C - 26.5°C = 2.7°C; percentage of hydrogen (H) = 0.7%; latent heat of steam = 587 cal / g.

$$\therefore \text{Gross calorific value} = \frac{(W + w)(t_2 - t_1)}{x} = \frac{(3,500 + 385)}{0.83} = 12,638 \text{ cal/g}$$

$$\text{Net calorific value} = [\text{GCV} - 0.09H \times 587] = (12,638 - 0.09 \times 0.7 \times 587) \text{ cal/g}$$

$$= (12,638 - 37) \text{ cal/g} = 12,601 \text{ cal/g}$$

Example-3: A gas used in an internal combustion engine had the following composition by volume: H₂ = 45%, CH₄ = 36%, CO = 15%, N₂ = 4%. Find the volume of air required for the combustion of 1 m³ of the gas.

Solution: 1 m³ of the gas contains H₂ = 0.45 m³; CH₄ = 0.36 m³; CO = 0.15 m³; N₂ = 0.04 m³.

Combustion reaction	Volume of O ₂ required (m ³)
H ₂ + 0.5O ₂ → H ₂ O	0.45 × 0.5 = 0.225
CH ₄ + 2O ₂ → CO ₂ + 2H ₂ O	0.36 × 2 = 0.720
CO + 0.5O ₂ → CO ₂	0.15 × 0.5 = 0.075
	Total = 1.020

$$\therefore \text{Volume of air required per m}^3 \text{ of the gas} = 1.020 \text{ m}^3 \times \left(\frac{100}{21} \right) = 4.857 \text{ m}^3$$

Example-4: Calculate the mass of air need for complete combustion of 5 kg of coal contain: C = 80%, H = 15%, O = rest.

Solution: 5 kg of coal contains: C = 4 kg; H = 0.75 kg; O = (5 – 4 – 0.75) kg = 0.25 kg.

∴ Amount of air required for complete combustion of 5 kg coal

$$\begin{aligned} &= \left[5 \times \left(\frac{32}{12} \right) + 0.75 \times \left(\frac{16}{2} \right) - 0.25 \right] \text{ kg} \times \left(\frac{100}{23} \right) \\ &= [13.333 + 6.000 - 0.25] \text{ kg} \times \left(\frac{100}{23} \right) = 82.97 \text{ kg} \end{aligned}$$

Example-5: A sample of coal contains: C = 93%; H = 6% and ash = 1%. The following data were obtained when the above coal was tested in bomb calorimeter:

(i) Weight of coal burnt = 0.92 g

(ii) Weight of water taken = 550 g

(iii) Water equivalent of bomb and calorimeter = 2,200 g

(iv) Rise in temperature = 2.42°C

(v) Fuse wire correction = 10.0 cal

(vi) Acid correction = 50.0 cal

Calculate gross and net calorific value of the coal, assuming the latent heat of condensation of steam as 580 cal/g.

Solution: Weight of coal sample (x) = 0.92 g; Weight of water (W) = 550 g; water equivalent of calorimeter (w) = 2,200 g; temperature rise ($t_2 - t_1$) = 2.42°C; acid correction = 50.0 cal; fuse wire correction = 10.0 cal; latent heat of steam = 580 cal / g; percentage of H = 6%.

$$\begin{aligned} \therefore \text{GCV} &= \frac{(W + w)(t_2 - t_1) - [\text{Acid} + \text{fuse corrections}]}{x} \\ &= \frac{(550 + 2,200) \times 2.42 - [50 + 10] \text{ cal}}{0.92 \text{ g}} = 7,168.5 \text{ cal/g} \end{aligned}$$

$$\begin{aligned} \therefore \text{NCV} &= [\text{GCV} - 0.09\text{H} \times \text{Latent heat of steam}] \\ &= (7,168.5 - 0.09 \times 6 \times 580) \text{ cal/g} = 6,855.3 \text{ cal/g} \end{aligned}$$

Example-6: A sample of coal was found to contain: C = 80%; H = 5%; O = 1%; N = 2%; remaining being ash. Calculate the amount of minimum air required for complete combustion of 1 kg of coal sample.

Solution: 1 kg of coal contains: C = 800 g; H = 50 g; O = 10 g; N = 20 g.

Combustion reaction	Weight of air required
$\begin{array}{ccc} \text{C} + \text{O}_2 & \longrightarrow & \text{CO}_2 \\ 12 & 32 & \end{array}$	$800 \text{ g} \left(\frac{32}{12} \right) = 2,146 \text{ g}$
$\begin{array}{ccc} 2\text{H} + 0.5\text{O}_2 & \longrightarrow & \text{H}_2\text{O} \\ 2 & 16 & \end{array}$	$50 \text{ g} \times \left(\frac{16}{2} \right) = 400 \text{ g}$
	Total = 2,546 g
	Less O in fuel = 10 g
	Net O ₂ required = 2,536 g

$$\therefore \text{Weight of air required} = 2,536 \text{ g} \times \left(\frac{100}{23} \right) = 11,026 \text{ g} = 11.026 \text{ kg}$$

Example-7: A gaseous fuel has the following composition by volume. Methane = 5%; hydrogen = 20%, carbon monoxide = 25%, carbon dioxide = 6%, and rest nitrogen. If 20% excess air is used for combustion, then calculate volume of air supplied per m³ of fuel and composition of dry flue gases.

Solution: 1 m³ of gaseous fuel contains: CH₄ = 0.05 m³; H₂ = 0.20 m³; CO = 0.25 m³; CO₂ = 0.06 m³ and N₂ = 0.46 m³ (by difference).

Combustion reaction	Volume of O ₂ required (m ³)	Volume of dry product (m ³)
CH ₄ + 2O ₂ → CO ₂ + H ₂ O	0.05 × 2 = 0.100	CO ₂ = 0.05 × 1 = 0.5
H ₂ + 0.5O ₂ → H ₂ O	0.25 × 0.5 = 0.100	—
CO + 0.5O ₂ → CO ₂	0.25 × 0.5 = 0.125	CO ₂ = 0.25 × 1 = 0.25
		CO ₂ = 0.06 From fuel
		N ₂ = 0.45
	Total = 0.325	CO ₂ = 0.36; N ₂ = 0.46

∴ Volume of air required (using 20% excess) per m³ of gaseous fuel

$$= 0.325 \text{ m}^3 \times \left(\frac{100}{21} \right) \times \left(\frac{120}{100} \right) = 1.857 \text{ m}^3.$$

Calculation of dry products of combustion:

$$\text{CO}_2 = 0.36 \text{ m}^3; \text{O}_2 = 20\% \text{ of } 0.325 \text{ m}^3 = \left(\frac{20}{100} \right) \times 0.325 \text{ m}^3 = 0.065 \text{ m}^3$$

$$N_2 = 0.46 \text{ m}^3 \text{ (of gaseous fuel)} + 1.857 \text{ m}^3 \left(\frac{77}{100} \right) \text{ of air} = 1.927 \text{ m}^3$$

$$\text{Total volume of dry products} = (0.36 + 0.065 + 1.927) \text{ m}^3 = 2.352 \text{ m}^3$$

$$CO_2 = \frac{0.36 \times 100}{2.352} = 15.306\%; N_2 = \frac{1.927 \times 100}{2.352} = 81.93\%; O_2 = \frac{0.065 \times 100}{2.352} = 2.764\%$$

Example-8: A sample of coal was found to have the following percentage composition: C = 54; H = 6.5; O = 6.0; N = 1.8; S = 3.2, moisture = 17.3 and reminder is ash. This coal on combustion with excess air gave 20.00 kg of dry product (flue gases) per kg of coal burnt. Calculate the percentage excess air used for combustion of coal.

Solution: 1 kg of coal contains: C = 540 g, H = 65 g; O = 60 g; N = 18; S = 32 g.

Combustion reaction	Weight of O ₂ required	Weight of dry flue gas
$C + O_2 \longrightarrow CO_2$ $H_2 + 0.5O_2 \longrightarrow H_2O$ $S + O_2 \longrightarrow SO_2$ –	$540 \times \left(\frac{32}{16} \right) = 1,440 \text{ g}$ $65 \times \left(\frac{16}{2} \right) = 520 \text{ g}$ $32 \times \left(\frac{32}{32} \right) = 32 \text{ g}$ –	$CO_2 = 540 \times \left(\frac{44}{12} \right) = 1,980 \text{ g}$ – $SO_2 = 32 \times \left(\frac{64}{32} \right) = 64 \text{ g}$ $N_2 \text{ (in coal)} = 18 \text{ g}$
–	Total = 1,992 g Less O ₂ in fuel = 60 g Net O ₂ required = 1,937 g	Total = 2,062 g
Corresponding weight of air = $1,937 \text{ g} \times \left(\frac{100}{23} \right) = 8,400 \text{ g or } 8.40 \text{ kg}$		

$$\text{Total weight of dry products} = \left[2,062 + 8,400 \times \left(\frac{77}{100} \right) \right] \text{ g} = 8,530 \text{ g or } 8.530 \text{ kg}$$

Now weight of flue gas per kg coal burnt = 20 kg

$$\therefore \text{Weight of excess air used} = (20 - 8.530) \text{ kg} = 11.47 \text{ kg}$$

$$\text{Hence, excess air used} = \frac{11.47 \times 100}{8.40} = 136.5\%$$

Questions:

1. Explain carbonisation of coal.
2. Describe in brief, the manufacture of metallurgical coke by Otto Hoffman's oven method.

3. Define octane number and cetane number. What are the structural features of hydrocarbons in unleaded petrol and diesel?
4. Discuss the importance of ultimate analysis of coal.
5. Explain carbonisation of coal.
6. Distinguish between gross and net calorific value of a fuel.
7. A gaseous fuel has the following composition by volume: H_2 = 25%, methane = 30%, ethane = 11%, ethene = 4.5%, butene = 2.5%, carbon monoxide = 6.0%, CO_2 = 8%, O_2 = 2% and N_2 = 12%. Calculate the air-fuel ratio and volumetric analysis of dry products of combustion using 40% excess air.
8. Calculate the minimum weight of air required for complete combustion of 1 kg of fuel containing: C = 90%, H = 3.5%, O = 3.0%, S = 0.5%, H_2O = 1.0%, N = 0.5% and ash = rest.
9. A sample of coal containing 92% C, 5% H, 3% ash. When this coal was tested in the laboratory for its calorific value in the bomb calorimeter, the following data were obtained:

 Weight of coal burnt = 0.95 g
 Weight of water taken = 700 g
 Water equivalent of bomb and calorimeter = 2,000 g
 Rise in temperature = 2.48°C
 Cooling correction = 0.02°C
 Fuse wire correction = 10.0 cal
 Acid correction = 60.0 cal
 Calculate the net and gross calorific values of the coal in cal / g. (Assume the latent heat of condensation of steam as 580 cal / g.).
10. Calculate the higher and lower calorific values of a coal sample containing 84% carbon, 1.5% sulphur, 0.6% nitrogen, 5.5% hydrogen and 8.4% oxygen.
11. Calculate the gross and net calorific value of a sample of coal having following composition: C = 80%, H = 7%, O = 3%, S = 3.5%, N = 2.1% and ash = 4.4%.

Text Book:

1. A Text book of Engineering, Chemistry, P. C. Jain and Monika Jain, 16 th Edition 2013, Dhanpat Rai Publishing Company (P) LTD, New
2. Engineering Chemistry O. G. Palanna 1 st Edition 2011, Mc Graw Hill, Education (India), Private Limited, Chennai.
3. A Text book of Engineering, Chemistry S. S. Dara 1 st Edition 2010, S. Chand & Company Ltd., New Delhi



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